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Python Modules

A Python module is a file containing Python definitions and statements. A module can define functions, classes, and variables. A module can also include runnable code. Grouping related code into a module makes the code easier to understand and use. It also makes the code logically organized.

Example: create a simple module

```
# A simple module, calc.py  
def add(x, y):  
    return (x+y)  
def subtract(x, y):  
    return (x-y)
```

Import Module in Python – Import statement

We can import the functions, classes defined in a module to another module using the import statement in some other Python source file.

Syntax:

```
import module
```

When the interpreter encounters an import statement, it imports the module if the module is present in the search path. A search path is a list of directories that the interpreter searches for importing a module. For example, to import the module calc.py, we need to put the following command at the top of the script.

Note: This does not import the functions or classes directly instead imports the module only. To access the functions inside the module the dot(.) operator is used.

Example: Importing modules in Python



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```
# importing module calc.py
import calc

print(calc.add(10, 2))
```

The from import Statement

Python's *from* statement lets you import specific attributes from a module without importing the module as a whole.

Example: Importing specific attributes from the module

```
from math import sqrt, factorial

# if we simply do "import math", then math.sqrt(16) and math.factorial() are
required.

print(sqrt(16))
print(factorial(6))
```

Import all Names – From import * Statement

The * symbol used with the from import statement is used to import all the names from a module to a current namespace.

Syntax:

```
from module_name import *
```

The use of * has its advantages and disadvantages. If you know exactly what you will be needing from the module, it is not recommended to use *, else do so.



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Example: Importing all names

```
from math import *  
# if we simply do "import math", then math.sqrt(16) and math.factorial() are  
required.  
print(sqrt(16))  
print(factorial(6))
```

Locating Modules

Whenever a module is imported in Python the interpreter looks for several locations. First, it will check for the built-in module, if not found then it looks for a list of directories defined in the sys.path. Python interpreter searches for the module in the following manner –

- First, it searches for the module in the current directory.
- If the module isn't found in the current directory, Python then searches each directory in the shell variable **PYTHONPATH**. The PYTHONPATH is an environment variable, consisting of a list of directories.
- If that also fails python checks the installation-dependent list of directories configured at the time Python is installed.

Example: Directories List for Modules

```
# importing sys module  
import sys  
# importing sys.path  
print(sys.path)
```

Importing and renaming module

We can rename the module while importing it using the as keyword.



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Example: Renaming the module

```
import math as gfg  
  
# if we simply do "import math", then math.sqrt(16) and math.factorial() are  
required.  
  
print(gfg.sqrt(16))  
print(gfg.factorial(6))
```

The dir() function

The dir() built-in function returns a sorted list of strings containing the names defined by a module. The list contains the names of all the modules, variables, and functions that are defined in a module.

```
# Import built-in module random  
import random  
print(dir(random))
```



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